age technologies to digital-age technologies will clearly be winners while economies that ignore digital technologies or are inefficient converters will lose out.

India's Digital Transformation Imperatives

Digital technologies are transforming economies the world over. The global digital economy is estimated to be worth about $5 trillion now, or about 5–6 per cent of global GDP. Indian technology entrepreneurs did well in starting and establishing IT services in India beginning in the 1990s and India today has a share of over one-third in global GDP of IT services of about $500 billion. However, India did not do well in electronics manufacturing and building software and the new-age businesses of cloud computing, IT-enabled services and services based on AI.

New digital start-ups have begun building India as a global force in software as a service (SaaS) and some major corporates have signalled their intention to get into semiconductors and chip making.

India's digital transformation agenda is fourfold.

First, the unique differentiating factor between digitally advanced and digitally backward economies is the ability to produce chips. The production of chips requires semiconductors and the ability to design chips. India is literally bereft of the capacity to produce semiconductors and has acquired very limited capacity to design and produce chips. This needs to change and India should aim at becoming a major semiconductors and chip-making and designing power.

Second, digital technology and infrastructure are developing and changing rapidly. With the advancement of machine learning and machine communications (IoT), the digital technology and infrastructure of computer networks, data storage, including cloud and data analytics, are providing amazing productivity, user experience and competitiveness to digitally advanced economies. While data storage and spectrum-based